AITOGY-GNSS-BASE-ESP-UBX

H ng d n k thu t n i b | Firmware V2.3.5 | C p nh t 25/10/2026

1. Ph m vi v nguy n t c v n h nh

B AITOGY-BASE-ESP nh n d li u t receiver GNSS, chuy n ti p RTCM l n caster b ng 4G ho c Wi-Fi v nh n i u khi n t h th ng CORS. Anten base ph i ng y n khi ph t c i ch nh. Sau khi di chuy n anten, c u h nh l i t a base tr c khi cho rover s d ng.

T i li u m t firmware **V2.3.5**, c p nh t **25/10/2026**. K t qu bi n d ch v ki m tra ph n m m kh ng thay th nghi m thu ngo i th c a. Kh ng cam k t th€i gian kh•i ng c nh hay ch nh x c o khi ch a ki m tra receiver, anten, m ng c i ch nh v h t a .

Ch	Khi s d ng	i u ki n ho n th nh
FIXED_LLA	,ƒ c„ t a i m t anten ng tin c y.	G i c u h nh fixed, ki m tra t a receiver v lu…ng RTCM.
SURVEY_IN	C n receiver t kh o s t v tr ng y n.	Receiver k t th†c kh o s t; g i l nh th nh c ng kh ng c„ ngh‡a kh o s t ƒ xong.
AUTO_CORS	C„ t i kho n rover nh n c i ch nh t CORS.	Chuy n rover, nh n c i ch nh, l y m^u RTK_FIXED, t l i base v ki m tra ph t RTCM.

, cao ph i d‰ng †ng quy c cŠa t a cung c p. V i c u h nh fixed theo LLA, s d ng cao ellipsoid. Kh ng thay tr c ti p b ng cao thŠy chu‹n ho c bŒ qua chi u cao anten/ i m quy chi u.

2. K t n i v chu n b

ng t n hi u TDM240x	K t n i
GPIO18 / RX GNSS cŠa ESP32	Nh n t TX cŠa receiver GNSS.
GPIO19 / TX GNSS cŠa ESP32	G i v o RX cŠa receiver GNSS.
GND	Chung mass gi a ESP32, receiver v b ngu…n theo thi t k b thi t b .
GPIO16 / RX modem	Nh n t TX modem.
GPIO17 / TX modem	G i v o RX modem.
Serial console	115200 baud; g i m t d•ng l nh v k t th†c b ng LF.
UART GNSS	M c nh 115200 baud, 8N1.

, i chi u nhƒn cŽng v s• … cŠa †ng board tr c khi u d•y. Kh ng suy ra ch•n c‘ m hay m c ngu…n t h nh nh cŠa board kh c. G‘n anten GNSS †ng cŽng, t n•i tho ng tr€i, c nh c• kh ; ki m tra SIM/APN v anten 4G tr c khi ch y auto base.

B n u-blox d‰ng receiver h ' tr " c c l nh c u h nh m firmware t o, m c ti u l d•ng ZED-F9P ph‰ h "p. N i UART1 receiver t i RX18/TX19 ESP32. M t s receiver/phi n b n ch" h ' tr " m t ph n CFG-VALSET ho c TMODE3; kh ng suy lu n m i receiver u-blox u t •ng th ch.

3. C u h nh m c nh v th ng tin c n b n giao

Th ng s	Gi tr / ngh a trong firmware
K t n i m ng	4G; c„ th chuy n Wi-Fi b ng ATG.
APN	v-internet; thay theo SIM th c t .
MQTT broker m c nh	45.117.179.134:1883; ph i kh p h th ng CORS ang tri n khai.
Backend WebSocket	base.aitogy.com:8000, giao th c ws; c n endpoint thi t b th c s truy c p "c.
Caster d ph•ng trong c u h nh m c nh	aitogy.com.vn:2101, mountpoint /test; kh ng ph i th ng tin c p ph t cho m i kh ch h ng.
K nh qu n l• ch nh	Topic theo serial esp_... do thi t b t o; qu n tr vi n c p quy n tr n CORS.
Tr ng th i backend	Chu k– m c nh 5 gi•y; k t n i m ng c„ th l m ch m vi c g i.
Health ki u c–	Chu k– m c nh 30 gi•y.

, •n v tri n khai b n giao: serial, t n tr m, lo i GNSS, caster/mountpoint ph t, t i kho n c i ch nh u v o n u d‰ng AUTO_CORS, APN ho c Wi-Fi v ng €i qu n l• tr n CORS. M t kh‹u/token do qu n tr vi n c p ri ng, kh ng l y t v d trong t i li u.

4. a thi t b l n CORS

1. C p ngu…n Žn nh, m• console 115200 baud v ghi l i serial esp_…, phi n b n firmware, lo i chip, baud nh n di n "c.
2. C u h nh m ng, MQTT v backend theo tri n khai. C c chu ' i trong d u <…> l gi tr c n thay, kh ng nh p nguy n d u ngo c.
3. L u th ng tin tr c, g i l nh ch n m ng sau c‰ng v l nh „ kh•i ng l i thi t b .
4. Tr n CORS, qu n tr vi n nh n/c p thi t b , t t n, tri n khai license v ch n †ng lo i chip.
5. Tri n khai d ch v caster: server 1, server 2 n u c n, mountpoint, t i kho n ph t v ch lu n ph t/theo y u c u.
6. Ch n FIXED_LLA, SURVEY_IN ho c AUTO_CORS. Theo d˜i k t qu l nh v ti n , sau „ ki m tra lu…ng RTCM b ng rover c l p.

V d c u h nh 4G:

```
ATG ESP SET 4G APN <apn>
ATG MQTT SET SERVER <mqtt_host>
ATG MQTT SET PORT 1883
ATG MQTT SET USER <mqtt_user>
ATG MQTT SET PASS <mqtt_password>
ATG ESP SET BACKEND HOST <backend_host>
ATG ESP SET BACKEND PORT 8000
ATG ESP SET CONNECTION 4G
```

V d Ži sang Wi-Fi:

```
ATG ESP SET WIFI SSID <ssid>
ATG ESP SET WIFI PASS <wifi_password>
ATG ESP SET CONNECTION WIFI
```

Console ATG t ch tham s theo d u c ch, kh ng h' tr" chu'i c„ kho ng tr'ng ho c gi tr r'ng. N u SSID/m t kh‹u c n d ng n y, li n h k™ thu t; kh ng d‰ng d u nh y gi l p chu'i.

Thi t b ch a "c c p quy n ho c ch a c„ license vˆn c„ th b o tr ng th i nh ng kh ng Š i u ki n ph t RTCM. Tr ng th i online cŠa k nh qu n l• c–ng ch a ch ng minh rover nh n "c c i ch nh.

5. Ch n ch base v x c nh n k t qu€

5.1 T a c nh

Nh p v‡ /kinh d ng th p ph•n v cao ellipsoid t nh b ng mšt. Ki m tra †ng i m quy chi u anten v h t a . Sau khi c u h nh, i chi u t a c t receiver, s hi n di n cŠa b n tin t a base v t a rover t i i m ki m tra.

5.2 Survey-in

, t anten ng y n, nh p th€i gian kh o s t theo gi•y. Nh nh u-blox c u h nh TMODE3 v i c th€i gian t i thi u v gi i h n ch nh x c. Receiver t quy t nh k t th†c survey theo i u ki n c u h nh; theo d˜i tr ng th i th c t sau khi g i.

Survey-in kh ng t t o ra t a tuy t i t ch nh x c kh o s t ch" v ch n ng ›ng nh€. Khi c n t a g'n v i h quy chi u c th , d‰ng i m *f* bi t ho c AUTO_CORS v th c hi n ki m tra c l p.

5.3 Auto base t• CORS

Firmware chuy n receiver sang rover, nh n RTCM t mountpoint c i ch nh v ch" ch p nh n GGA m i v i tr ng th i **RTK_FIXED**. RTK_FLOAT kh ng "c d‰ng l m mˆu auto base.

Lu…ng m c nh c n 5 l n ki m tra FIX li n ti p theo nh p kho ng 1 gi•y, sau „ l y t i a 60 mˆu trong c a sŽ 60 gi•y. C n t nh t 15 mˆu h"p l ti p t c; kh ng m c nh coi k t qu lu n c„ Š 60 mˆu. Thi t b t l i t a base r…i kh i ph c d ch v ph t *f* c u h nh.

Th€i gian ch€ FIX m c nh 3600 gi•y; qu n tr vi n c„ th c u h nh theo nghi p v . Khi m t FIX, m t k t n i, thi u mˆu ho c c„ l nh d ng, xem l• do trong ti n workflow. Vi c kh i ph c k t n i caster sau th t b i kh ng t ch ng minh receiver *f* tr• v c u h nh base mong mu n.

Ph n chuy n VN2000 hi n d‰ng tham s t ch h"p v phšp chi u theo l a ch n t"nh/kinh tuy n. Firmware kh ng gi i m*f* c c b n tin datum RTCM 1021/1023/1025. Ph i i chi u m c ki m tra th c a; kh ng coi nh*f*n VN2000 l ch ng nh n †ng h t a .

6. Theo d,i tr ng th i v lu*f*ng d„ li u

Hi…n th /tr ng	C ch hi…u
awaiting_license / unprovisioned	Ch a Š i u ki n c p quy n/license ph t RTCM.
locked	Thi t b b kh„a to n b ; li n h qu n tr vi n.
sleep	, ang ch€ k ch ho t lu…ng theo y u c u; ch a ch'c l m t m ng.
online	K nh qu n l• ho t ng; c n ki m tra th m t ng server NTRIP.
auto_setup_* / reference_check*	Workflow ang th c hi n m t b c c u h nh ho c ki m tra.
last_command_result	K t qu x l• l nh; l nh kh•i ng workflow th nh c ng ch" ngh‡a *f* nh n vi c.
auto_base_progress	Theo d~i phase, s m^u v l• do completed/failed/stopped cŠa workflow.
server1/2_connected v server1/2_bps	K t n i v s byte/gi•y g i t i t ng caster.
rtcm_input_bps	S byte RTCM/gi•y "c „ng khung • UART; kh ng ph i ch ng nh n CRC h"p l .
GGA/fix/satellites	Ch t l "ng nghi m GNSS v d li u v tr ang nh n.

V d c u tr†c health ki u c– (gi tr minh h a):

```
{
  "uptime_s": 120,
  "free_heap_bytes": 120000,
  "connected_via": "GSM",
  "rssi_dbm": 20,
  "mqtt_ok": true,
  "ntrip_ok": true,
  "gnss_data_ok": true,
  "rtcm_decoder": false,
  "rtcm_types": {}
}
```

connected_via d‰ng GSM ho c WiFi. gnss_data_ok ph n nh *f* nh n khung RTCM g n •y, trong ng ›ng 5 gi•y cŠa t c v health. rtcm_types r'ng v ESP32 kh ng m t ng lo i b n tin. rssi_dbm l t n tr €ng c–: Wi-Fi d‰ng RSSI, c•n 4G ang gi gi tr ch t l "ng t n hi u th do modem tr v , ch a t chuy n sang dBm. Kh ng d‰ng gi tr „ nh s dBm 4G *f* hi u chu‹n.

H th ng nh n NMEA GGA/GSA/GSV/GST theo d~i GNSS. ESP32 kh ng m c nh t o m t b n sao to n b RTCM l n MQTT. C c tr €ng c„ t n valid_packets hi n l b m „ng khung, kh ng c„ ki m tra CRC RTCM. hacc c„ th l y t GST ho c fallback HDOP; kh ng coi m i gi tr hacc u l ch nh x c mšt *f* x c nh n.

7. Danh m c l nh ATG

ATG d‰ng c u h nh k t n i v h' tr" k™ thu t. œu ti n CORS c u h nh base v c k t qu l nh theo t ng thi t b . T kh„a d‰ng †ng ch hoa; m'i d•ng k t th†c b ng LF. L nh SET th €ng l u c u h nh, c n kh•i ng l i k t n i c l i gi tr m i.

7.1 ESP v m ng

C† ph p	T c d ng
ATG ESP RESTART	Kh•i ng l i ngay.
ATG ESP SET CONNECTION 4G	L u ch n 4G v kh•i ng l i.
ATG ESP SET CONNECTION WIFI	L u ch n Wi-Fi v kh•i ng l i.
ATG ESP SET WIFI SSID <ssid>	L u SSID kh ng c„ kho ng tr ‘ng.
ATG ESP SET WIFI PASS <password>	L u m t kh‹u kh ng c„ kho ng tr ‘ng.
ATG ESP SET 4G APN <apn>	L u APN.
ATG ESP SET 4G USER <user>	L u t n •ng nh p APN khi nh m ng y u c u.
ATG ESP SET 4G PASS <password>	L u m t kh‹u APN khi nh m ng y u c u.
ATG ESP SET BACKEND HOST <host>	L u hostname/IP cŠa endpoint WebSocket, kh ng kžm scheme ho c €ng dˆn.
ATG ESP SET BACKEND PORT <port>	L u cŽng WebSocket; ch p nh n 1-65535.

7.2 MQTT v NTRIP ph t

C† ph p	T c d ng
ATG MQTT SET SERVER <host>	L u broker MQTT.
ATG MQTT SET PORT <port>	L u cŽng MQTT; nh p s h“p l 1-65535.
ATG MQTT SET USER <user>	L u t n •ng nh p broker.
ATG MQTT SET PASS <password>	L u m t kh‹u broker.
ATG MQTT SET PUBTPCHEALTH <topic>	L u topic health ki u c–.
ATG MQTT SET SUBTPCCMD <topic>	L u topic subscribe ATG ki u c–; thay topic c n k™ thu t ki m tra callback th c nh n.
ATG NTRIP SET CSTRADDR <host>	L u caster m c nh/d ph•ng.
ATG NTRIP SET CSTRPORT <port>	L u cŽng caster; nh p s h“p l 1-65535.
ATG NTRIP SET MNTPNT <mountpoint>	L u mountpoint m c nh.
ATG NTRIP SET CSTRAUTH <password>	L u m t kh‹u source nguy n v•n; kh ng nh p base64 cŠa username:password.

D ch v *f* c p t CORS c„ th u ti n h•n c u h nh NTRIP m c nh l u b ng ATG. Mu n thay t i kho n/mountpoint ang v n h nh, ch”nh d ch v tr n CORS r…i tri n khai l i.

7.3 GNSS v t l i c u h nh

C† ph p	T c d ng
ATG GNSS ROVER	Chuy n receiver sang rover ch‹n o n; kh ng t •ng nh p t i kho n rover CORS.
ATG GNSS BASE SURVEY_IN <seconds> <accuracy_m>	G i batch survey-in theo lo i chip; m c nh ch" xu t RTCM, kh ng xu t NMEA diagnostic.
ATG GNSS BASE SURVEY_IN <seconds> <accuracy_m> W_DIAG	Survey-in v xu t c RTCM l^n c c b n tin NMEA diagnostic; W_DIAG kh ng ph•n bi t hoa th €ng.
ATG GNSS BASE FIXED <lat> <lon> <alt> <accuracy_m>	G i t a LLA c nh; cao t nh b ng mšt ellipsoid.
ATG GNSS BASE <message> <port> ON	B t b n tin RTCM tr n cŽng ch" nh.
ATG GNSS BASE <message> <port> OFF	T't b n tin RTCM tr n cŽng ch" nh.
ATG GNSS SET CHIP UBLOX	L u l a ch n chip v kh•i ng l i; boot v^n th c hi n nh n di n ph n c ng.
ATG CONFIG RESET	, t c€ n p l i m c nh v kh•i ng l i; c„ th l m Ži m ng v c u h nh GNSS.

CŽng l nh RTCM h' tr" I2C/UART1/UART2/USB/SPI; €ng n i ESP32 trong c u h nh chu‹n l UART1. V d : ATG GNSS BASE 1074 UART1 ON. B n tin ch a c„ trong b ng l nh firmware c„ th b t ch i d‰ receiver t n„ h' tr".

L nh ATG ki u c– chŠ y u tr log Serial, kh ng c„ k t qu JSON ri ng theo command_id nh l nh CORS. Kh ng coi d•ng Ÿ f g i l receiver f ch p nh n c u h nh. Reset c u h nh nh c c b kh ng thay th thao t c g› thi t b /thu h…i quy n tr n CORS; qu n tr vi n ph i x l• tr ng th i c p ph t khi c n.

8. RTCM, ephemeris v c u h nh receiver

8.1 B€n tin v gi i h n h‡ tr^

B ng l nh RTCM c c b hi n c„ 1005, 1074, 1084, 1094, 1124, 1077, 1087, 1097 v 1230. Nh nh u-blox ch a c„ b t o l nh ephemeris 1019/1020/1042/1044/1045 nh Unicore; kh ng qu ng b t nh n•ng „ ch" v firmware d‰ng chung ki u Command.

Survey/fixed d‰ng UBX-CFG-TMODE3; output d‰ng CFG-VALSET/CFG-MSG v l u CFG-CFG. Firmware nh tuy n c c output USB/UART2 trong batch backend v UART1 v t nh l i checksum. C n ki m tra receiver th c s h' tr" c c message/key "c c p.

8.2 Ki…m tra tr‰c ti p bŠng u-center

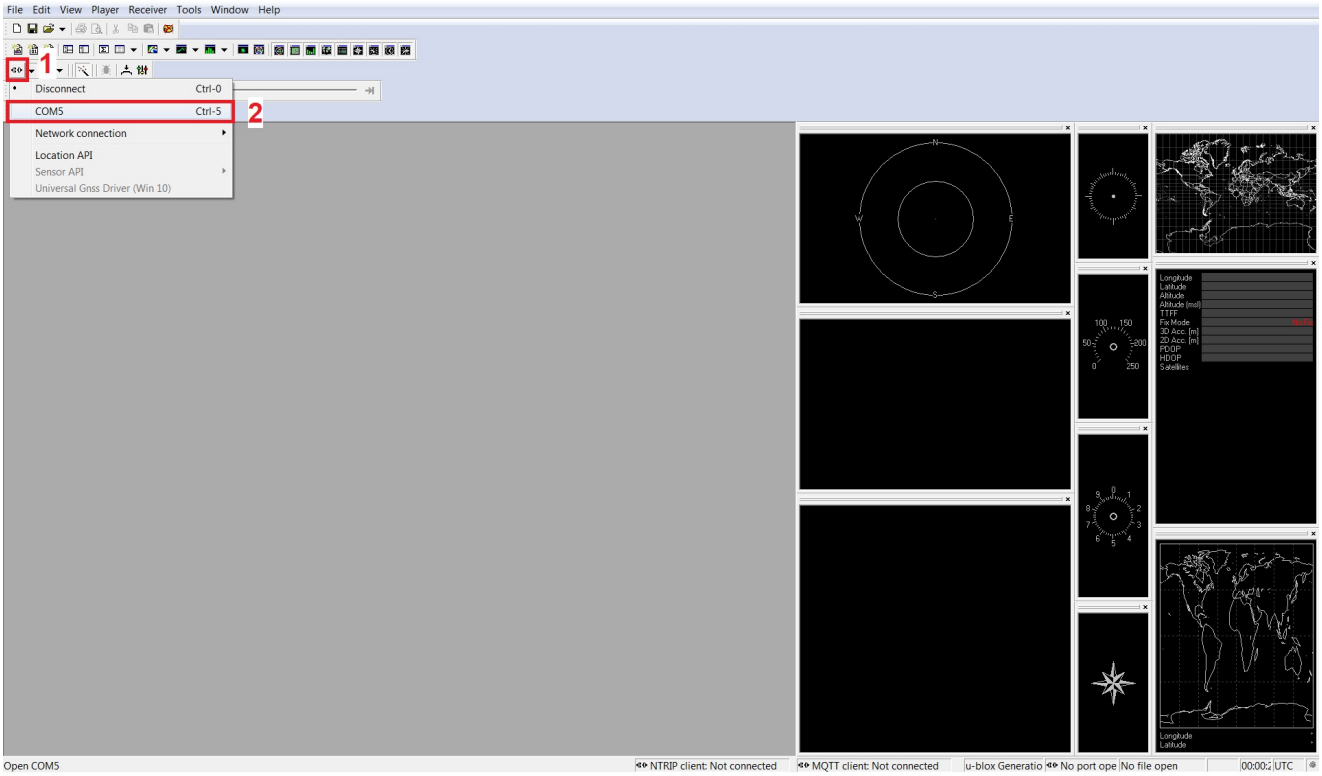
D‰ng c ng c c u h nh u-blox ph‰ h"p model v phi n b n receiver. B nh d i •y minh h a giao di n u-center cŽ i n f c„ trong d n; t n menu c„ th kh c • phi n b n kh c. K t n i †ng cŽng USB receiver, x c nh n model/version v baud tr c khi thay c u h nh.

1. Ch n k t n i receiver v d‰ng t d• baud khi ch a bi t t c hi n t i.
2. M• Packet Console/Message View ph•n bi t NMEA, UBX v RTCM; RTCM kh ng ph i v•n b n thu n.
3. C u h nh UART1 l cŽng n i ESP32, ki m tra baud 115200 v protocol input/output ph‰ h"p.
4. Trong c u h nh Time Mode/TMODE3, ch n survey-in ho c fixed v nh p †ng •n v .

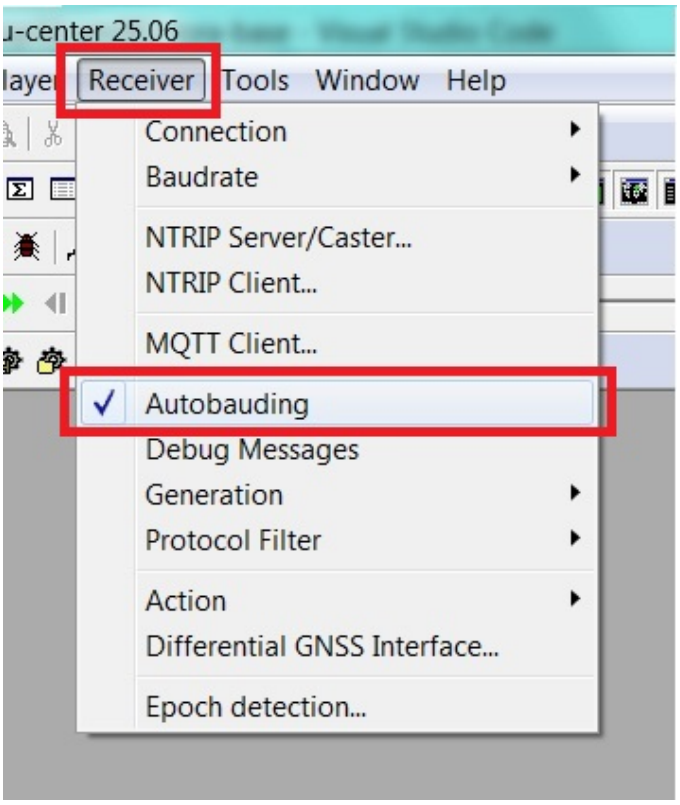
5. B t b n tin c n thi t • UART1, g i l nh, l u v o b nh receiver v ki m tra l i sau m t ngu…n.

Kh ng g i chu'i ASCII MODE BASE cŠa Unicore v o receiver u-blox. u-blox d‰ng g„i UBX nh ph•n v ki m tra ACK/NAK ph‰ h "p.

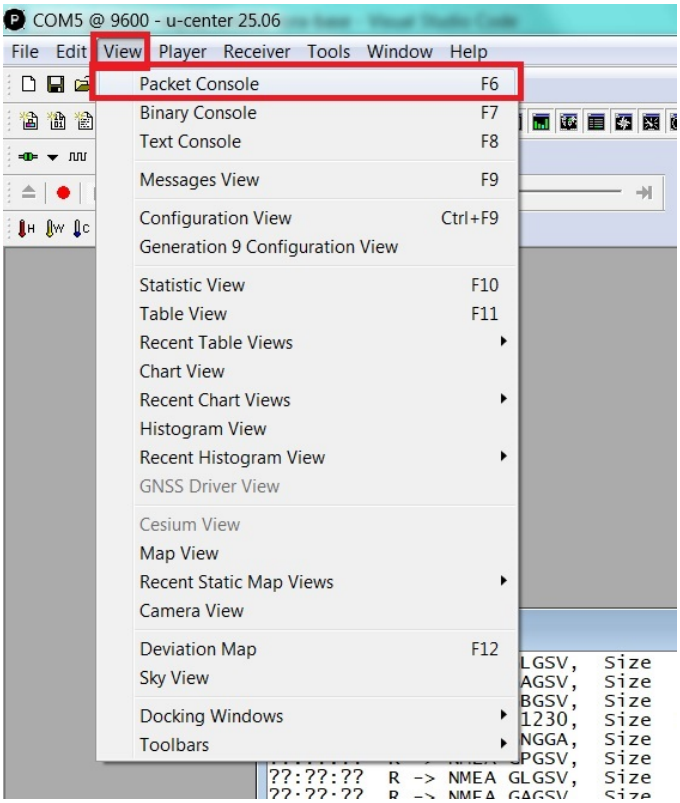
8.3 H nh minh h a u-center



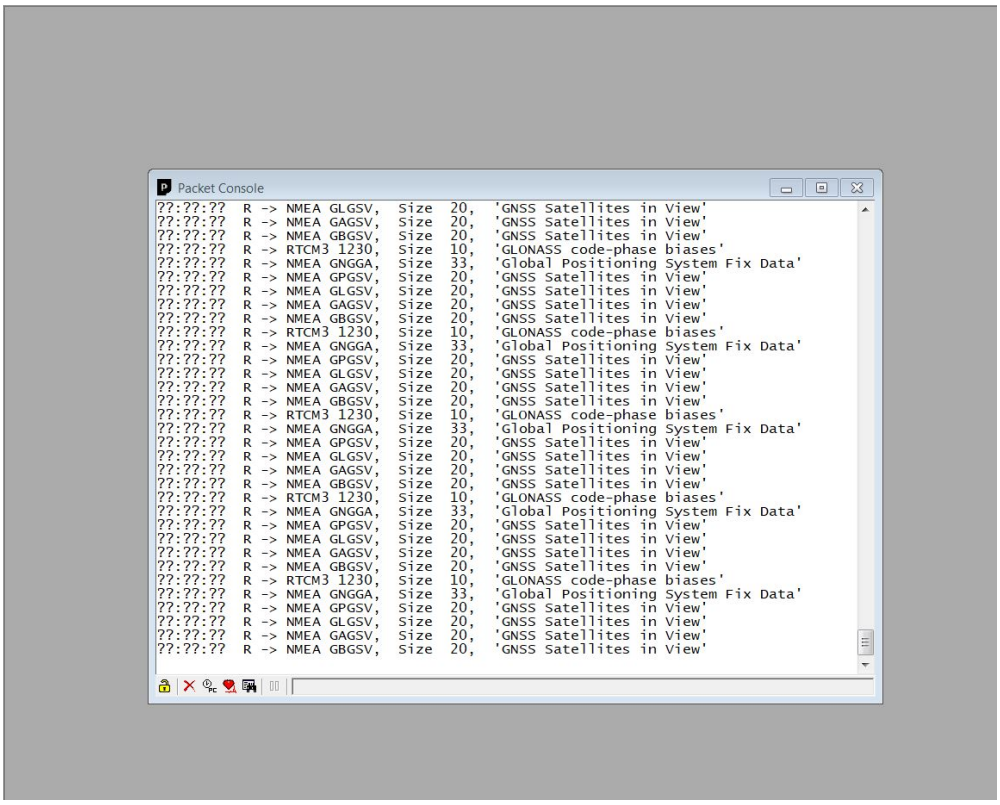
K t n i receiver trong giao di n u-center cŽ i n.



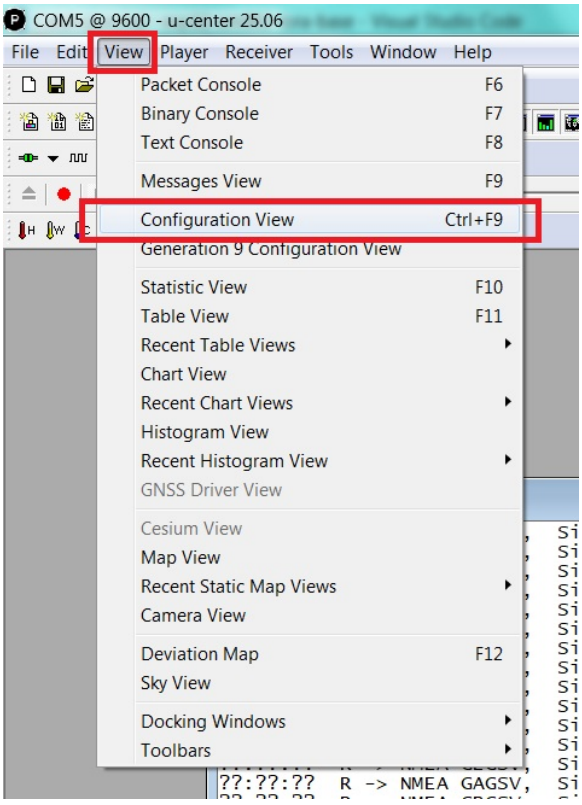
T d• baud cŠa k t n i receiver.



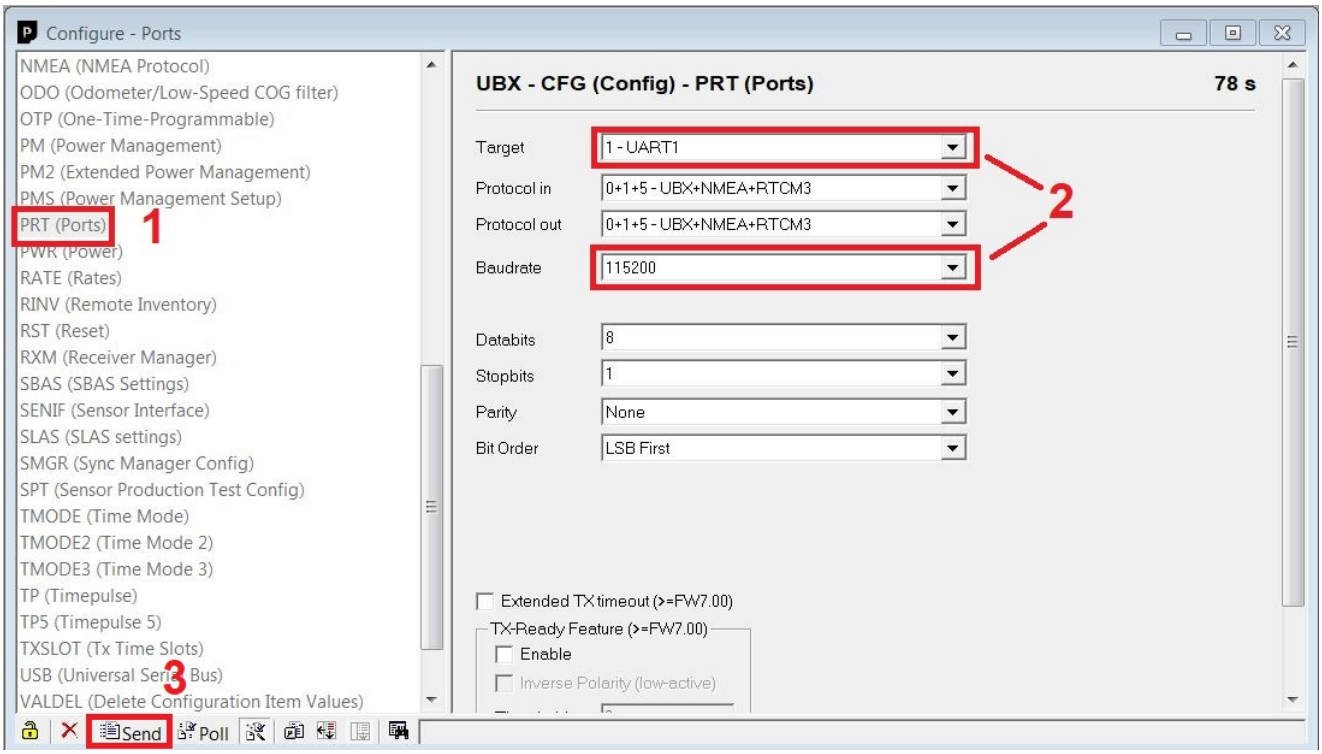
M• c a sŽ theo d˜i b n tin receiver.



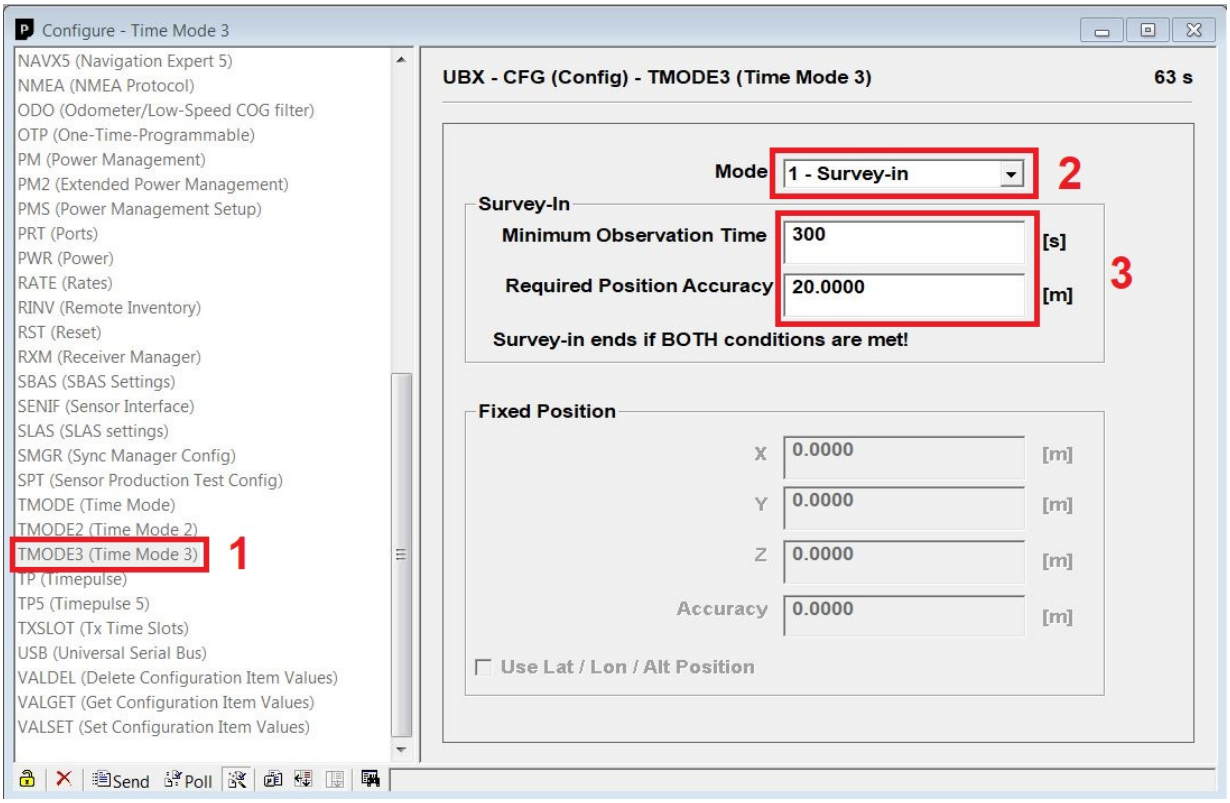
Packet Console hi n th c c lo i g„i nh n “c.



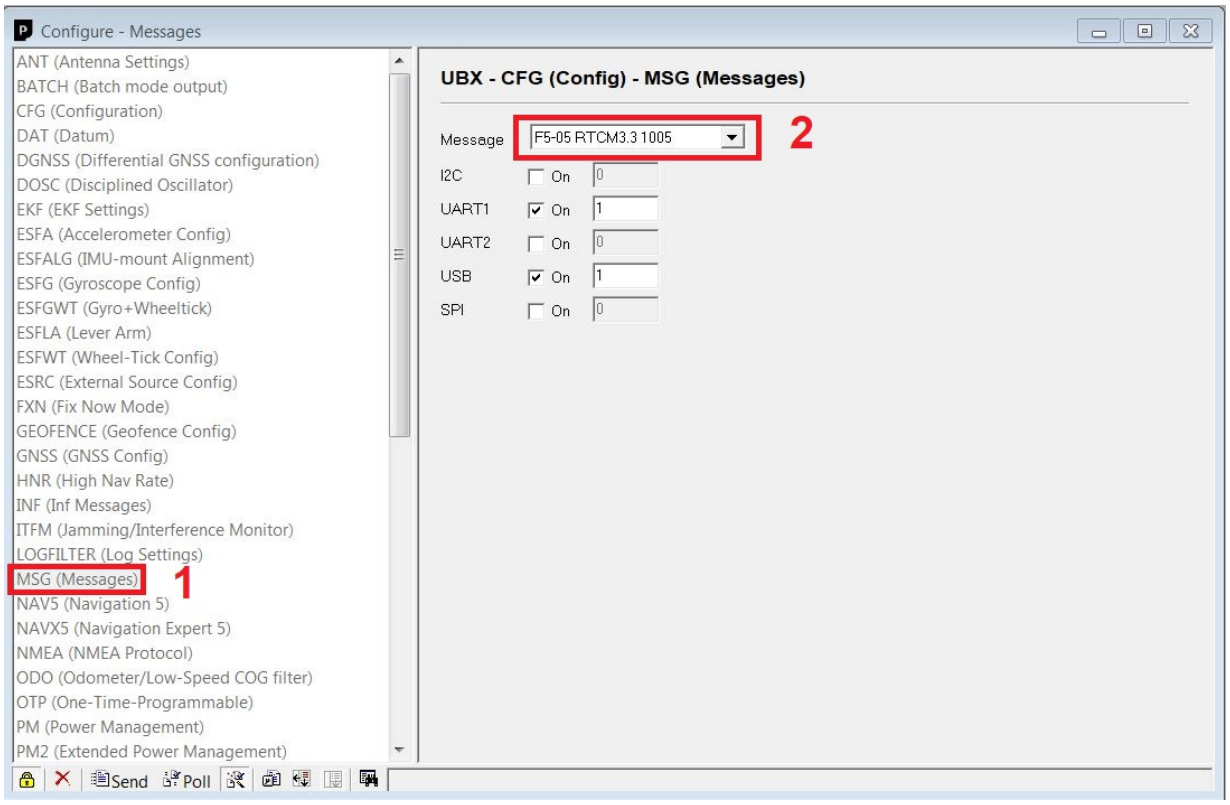
M• giao di n c u h nh receiver.



Ch n cŽng UART v baud cŠa receiver.



C u h nh Time Mode/survey-in.



C u h nh output b n tin tr n t ng cŽng.

9. X l s‰ c

D u hi u	Ki…m tra v h ng x l
Kh ng th y log console	, †ng cŽng USB/driver, 115200 baud, ngu…n v c p truy n d li u.
UNKNOWN/INVALID, kh ng c„ GGA	RX/TX u chšo, chung GND, †ng cŽng receiver, †ng baud; ki m tra tr c ti p receiver khi c n.
MQTT "c nh ng CORS ch a qu n l• "c	Broker/topic theo serial, thi t b ƒ nh n/c p v license; ki m tra command/result thay v ch" nh n k t n i TCP.
WebSocket kh ng k t n i	, †ng host/port endpoint thi t b ; firmware hi n d‰ng ws, kh ng ph i wss. MQTT vˆn l k nh i u khi n kh c.
C„ NMEA nh ng RTCM b ng 0	, †ng ch base, survey ƒ xong, †ng output UART v b n tin RTCM ƒ b t.
C„ RTCM UART nh ng caster kh ng nh n	, †ng source account/mountpoint/version NTRIP; server "c b t, kh ng kh„a v Š license/provision.
C„ hai k t n i c‰ng t i kho n rover	X c minh client n o o, client n o thu c d ch v monitor/decode tr n caster; kh ng x„a t i kho n khi ch a x c nh.
Auto base kh ng FIX	T i kho n rover, mountpoint u v o c„ d li u, GGA m i, cŽng RTCM input, anten/ che khu t v kho ng c ch tr m.
Auto base thi u mˆu	Ki m tra FIX c„ duy tr , GGA c„ m i v Š mˆu h"p l trong c a sŽ l y mˆu.
C u h nh GNSS b o l'i	, c ACK/NAK v CFG-TMODE3 readback khi chuy n rover; ki m tra phi n b n receiver h' tr" l nh ang d‰ng.
Kh•i ng l i vˆn d‰ng c u h nh c–	, i chi u service ang c p t CORS v i c u h nh ATG; ki m tra l u NVS v k t qu tri n khai d ch v .
T a l ch d‰ rover FIX	H quy chi u, cao, chi u cao anten, t a base v phšp chuy n VN2000; ki m tra tr n m c c l p.

Khi g i y u c u h' tr": cung c p serial, phi n b n, th€i i m l'i, phase/l• do workflow, lo i receiver, baud/cŽng n i, server n o l'i v log ƒ che m t kh‹u/token. Kh ng g i l i reset nhi u l n khi ch a x c nh l'i.

10. Nghi m thu v b€o tr

1. Ch p/ghi nh n †ng u d•y, ngu…n, anten, SIM/APN v serial "c c p.
2. T'‘t/b t ngu…n ki m tra thi t b n p †ng nh n d ng, d ch v , lo i GNSS v c u h nh base.
3. V i fixed/survey: i chi u t a base v b n tin tham chi u sau khi receiver Žn nh.
4. V i auto base: ki m tra t ng b c chuy n rover, nh n c i ch nh, RTK_FIXED, s mˆu, chuy n base v ph c h…i output.
5. D‰ng rover c l p k t n i †ng mountpoint, ki m tra tuŽi c i ch nh, FIX v i m m c ƒ bi t.
6. Ki m tra ri ng server 1/2, lu n ph t/theo y u c u v kh„a/m• t ng server; kh„a kh ng "c b fail-open bŒ qua.
7. Th m t m ng r…i ph c h…i; ki m tra k t qu th c t , kh ng ch" žn ho c nhƒn online.
8. Sau khi di chuy n anten, thay receiver/firmware ho c thay h t a , th c hi n l i ph n nghi m thu li n quan.

Kh ng c„ thao t c o ph n c ng t ng n o “c th c hi n ch” b•i vi c c ho c xu t b n t i li u n y. B n hi n t i ƒ i chi u code; nghi m thu ngo i hi n tr €ng do •n v tri n khai ghi nh n ri ng.

11. Ki n tr†c v ranh gi i giao th‹c

Firmware ESP32 l ch •ng tr nh C++, kh ng ch y Python agent. M t binary ch a hai b t o l nh, l a ch n theo lo i receiver nh n di n “c ho c c u h nh backend. Kh ng c•n macro GNSS_MODULE_TYPE ch n m t lo i chip khi bi n d ch.

Th nh ph n	Tr ch nhi m
gnss_cmd_builder	Ki u Command/CommandList, t‰y ch n v dispatch theo chip.
ublox_commands	T o g„i UBX, checksum, TMODE3, c u h nh output v nh tuy n sang UART1.
Ubx_Reply	T ch khung UBX c„ gi i h n b m v ki m tra checksum; b c Rover d‰ng ACK/NAK/readback.
RTCM receiver	T ch bi n NMEA/UBX/RTCM qua nhi u l n c UART, c p nh t telemetry; kh ng x c th c CRC RTCM.
Backend agent	Nh n l nh MQTT/WebSocket, quy n thi t b , workflow, l u tr ng th i v k t qu l nh.
NTRIP transport	Hai source stream v m t phi n rover t m d‰ng l i client server 1.

D li u v n h nh: receiver ph t RTCM qua UART v o ESP32; ESP32 ‹y l n caster. Khi auto base/reference check, chi u ng “c l caster c i ch nh ¡ phi n rover ¡ UART GNSS. NMEA GGA/GSA/GSV/GST i l n backend t nh tr ng th i; kh ng tri n khai decoder RTCM tr n ESP32.

11.1 UART, task v mutex

- TDM240x: GNSS RX18/TX19, modem RX16/TX17; nh nh BOARD_2A53N d‰ng GNSS RX22/TX23. Kh ng p d ng pin cŠa board n y cho board kia.
- GNSS RX ring 8192 byte; mutex timeout m c nh 1500 ms. D• boot l n l “t 115200, 38400, 9600, 19200, 57600, 230400, 460800 baud.
- task MQTT c•n x l• WebSocket v JSON; stack c u h nh 12288 byte. NTRIP 6144 byte, health v console 4096 byte m’i task.
- B “i ACK c u h nh chi m rtcmBufferMutex r…i gnssSerialMutex task c kh ng l y m t ph n h…i. tcpStreamMutex b o v thao t c m ng/modem.
- Mux modem: source 1/rover l 0, MQTT l 1, source 2 l 2, WebSocket l 3. D ng source tr c phi n rover t m.
- GGA g i upstream l y t receiver; gi i h n kho ng 5 gi•y/l n v kh ng g i b n ƒ c–. Kh ng t o t a gi duy tr FIX.

11.2 KhŒi t o v nh n di n

Boot n p identity/quy n/d ch v tr c khi m• transport, d• UBX-MON-VER ho c VERSION ch n chip. GNSS_MDL_TYPE d‰ng cho ATG, SENSOR_TYPE d‰ng cho backend: c n i chi u c hai khi ch‹n o n c u h nh l ch. ATG SET CHIP kh ng b o m šp model qua m t l n boot v boot l i d• ph n c ng.

Ch" l n u/reset m c nh m i g i bootstrap theo chip ƒ d•: UM980 c u h nh survey/output COM2 v ephemeris, u-blox c u h nh output UART1. Kh ng ph i m i l n kh•i ng u ghi l i to n b c u h nh GNSS. Sau nh n di n kh ng h "p l , kh ng coi thao t c kh•i t o l th nh c ng.

12. Giao th‹ c backend v i u ki n nh n l nh

Topic ch nh d‰ng serial th c:

```
pi/devices/<serial>/status
pi/devices/<serial>/raw_data
pi/devices/<serial>/command
pi/devices/<serial>/control_ack
pi/devices/<serial>/base_config_state
pi/devices/<serial>/service_config_state
```

C„ subscribe t •ng th ch /commands v pi/device/<serial>/command. WebSocket n i €ng dˆn /ws/pi/<serial>?token=<license> qua ws, g i status_update/nmea_update. MQTT hi n l TCP kh ng TLS; kh ng suy lu n c„ TLS t vi c website qu n tr d‰ng HTTPS.

L nh JSON c„ command, command_id v payload. original_config cŠa EXECUTE_RAW_COMMANDS n m • c p envelope, kh ng n m trong payload. V d d i •y ch" m t giao th c; g i qua dispatcher CORS p d ng quy n, command ID v ki m tra tham s .

```
{
  "command": "LOCK_DEVICE",
  "command_id": "example-lock-server2",
  "payload": {"locked_servers": [2]}
}
```

L nh	Tr ng ch nh v h nh vi
PROVISION_DEVICE	payload.name; t t n v nh d u thi t b ƒ c p ph t.
DEPLOY_LICENSE	payload.license_key; ki m tra nh d ng token dt_... ho c license s c–, l u th nh c ng m i b o nh n v reboot.
DEPLOY_SERVICE_CONFIG	Payload l object d ch v ; l u, n p l i endpoint v th k t n i.
EXECUTE_RAW_COMMANDS	payload.commands_b64; original_config c„ mode BASE, sensor_type v params n u c u h nh base.
SET_RTCM_STREAM_ACTIVE	payload.active; t‰y ch n server_id, mountpoint v address ch n lu…ng. Lu…ng lu n ph t bŒ qua y u c u t 't theo nhu c u.
TRIGGER_AUTO_BASE	ip, port, user, password, mountpoint, sensor_type; c c t‰y ch n timeout, province_code, l0_deg, k0, enable_itrf_vn2000_transform.
TRIGGER_BASE_REFERENCE_CHECK	K t n i rover, reference_coords v c u h nh base ƒ l u; o so s nh, sau „ c g 'ng kh i ph c base c–.
STOP_AUTO_BASE	Y u c u hŠy workflow; theo d˜i phase stopped/failed v k t qu cleanup.
APPLY_VN2000_PROVINCE	province_code v c u h nh t a base h "p l ; p d ng phšp chuy n t"nh/kinh tuy n.
LOCK_DEVICE	Payload r 'ng kh„a c thi t b ; locked_servers ch n server 1/2.

L nh	Tr ng ch nh v h nh vi
UNLOCK_DEVICE	X„a kh„a to n thi t b v kh„a server.
CHECK_BASE_STATUS	G i poll tr ng th i receiver; k t qu success ch" x c nh n y u c u truy v n ƒg i.
REBOOT_DEVICE	L u tr ng th i v kh•i ng l i.
DELETE_DEVICE	X„a tr ng th i thi t b theo handler v kh•i ng l i; ph i h"p v i qu n tr backend tr c khi c p l i.

L nh kh ng bi t b t ch i. Kh„a to n thi t b ch n l nh ngo i kh„a/m•. Ch a c„ license ch" cho phšp nh„m kh„a/m•, c p thi t b v tri n khai license. Kh ng "c coi vi c subscribe topic l ƒc„ quy n th c thi m i l nh.

K t qu c„ running, success, error ho c rejected g'n v i command_id; workflow ch y b t …ng b c„ ti n ri ng. Kh ng g p success cŠa vi c x p h ng workflow v i completed cŠa thao t c ngo i hi n tr €ng.

12.1 D ch v NTRIP v kh•a theo server

D ch v d‰ng server1_enabled/server2_enabled, serverhost1/2, port1/2, username1/2, password1/2, mountpoint1/2, ntrip_version1/2, server1/2_stream_on_demand v server1/2_stream_active. Ki u NTRIP v1 d‰ng SOURCE v i m t kh‹u nguy n v•n; v2 d‰ng POST v HTTP Basic do firmware t mƒ h„a.

```
{
  "server1_enabled": true,
  "serverhost1": "<caster_host>",
  "port1": 2101,
  "username1": "<source_user>",
  "password1": "<source_password>",
  "mountpoint1": "<station_mountpoint>",
  "ntrip_version1": 1,
  "server1_stream_on_demand": false,
  "server1_stream_active": true,
  "server2_enabled": false
}
```

LOCK_DEVICE v i [1,2] kh„a c hai output nh ng vˆn kh c kh„a to n thi t b . [] ho c UNLOCK_DEVICE bŒ kh„a; kh ng t b t endpoint b t'‘t. Khi m t ACK i u khi n kho ng 45 gi•y, fail-open c„ th m• lu…ng theo ch nh s ch l u; license, provision v kh„a vˆn l i u ki n b'‘t bu c. ACK m i h"p l c n ph c h…i Žn nh kho ng 5 gi•y tho t fail-open.

13. ACK, readback v m‹c b€o €m

B t o l nh u-blox ph t UBX nh ph•n, b c Ubx_Reply ki m tra d i v checksum. Trong applyRoverMode, firmware gi quy n UART, g i t ng l nh v quan s t ACK/NAK t i a 1500 ms. L'i ACK • c u h nh port t‰y ch n kh ng t d ng c b c Rover.

,i u ki n quy t nh l poll CFG-TMODE3 c l i mode 0, sau „ c„ GGA UART m i trong 10 gi•y. Kh ng g i b c chuy n rover l ƒx c nh n ch" t vi c ghi th nh c ng UART. Receiver kh c th h kh ng h' tr" TMODE3/readback n y c n "c nh gi ri ng.

C c batch c u h nh base u-blox c•n l i ch a ki m tra ACK/readback y Š: €ng ghi c„ th ch" x c nh n g i UART. , €ng EXECUTE_RAW_COMMANDS u-blox hi n ch a ki m tra Š s byte ghi t ng g„i. CHECK_BASE_STATUS ch" g i poll, task c GNSS ti p nh n byte v ; kh ng t b o m m t b n readback c u h nh y Š ƒ "c xu t l n tr ng th i.

13.1 Gi i h n ng ATG cŽ

sendGnssCommands cŠa ATG ghšp CommandList th nh m t dƒy byte r…i g i, c log t i a kho ng 500 ms; kh ng ki m tra ACK ri ng t ng l nh. C c marker $DELAY_...$ cŠa batch Unicore kh ng “c €ng ATG n y th c thi th nh kho ng ch€. Do „ d‰ng CORS cho c u h nh base Unicore c„ ki m tra ph n h…i, v kh ng k t lu n receiver ƒ nh n Š c u h nh t log ŸƒgicŠa ATG.

SUBTPCCMD “c l u v subscribe theo Preferences nh ng callback ATG vˆn so topic v i h ng TOPIC_SUB_CMD. ,Ži topic c„ th l m ATG kh ng “c dispatch; tri n khai m i n n d‰ng topic JSON theo serial. MQTT PORT/CSTRPORT chuy n s sang uint16_t m ch a c„ ki m tra mi n y Š; c ng c g i ph i ki m tra s 1-65535.

13.2 Auto base v reference check

Auto base ch” l y RTK_FIXED v i GGA m i; m c nh c n 5 l n ki m tra FIX li n ti p theo nh p kho ng 1 gi•y. C a sŽ l y mˆu 60 gi•y, m c ti u 60, t i thi u 15. Payload samples kh c 60 “c ghi nh n l overridden, kh ng thay m c ti u cŠa auto base. Reference check d‰ng samples ri ng, gi i h n 5-120, m c nh 60 v timeout m c nh 300 gi•y.

Khi reference check d ng/k t th†c, cleanup c g‘ng t l i t a base ƒ l u v ph c h…i transport. Auto base th t b i tr c b c t l i base ch” kh i ph c source transport, ch a lu n kh i ph c receiver mode/t a c–. Ki m tra l i receiver tr c khi a mountpoint v o khai th c.

VN2000 d‰ng tham s t ch h“p; kh ng c„ decoder datum t RTCM. T a NMEA g…m cao anten theo b n tin receiver; ki m tra quy Ži ellipsoid/geoid v chi u cao anten tr c khi s d ng k t qu nh t a m c.

14. L u c u h nh, c p nh t v kh i ph c

Nh•m d„ li u	V tr / c i…m
Wi-Fi, APN, MQTT, legacy NTRIP	Preferences myPrefs; m t s thay Ži ch” n p l i sau reboot.
Serial, t n, chip, provision, license, kh„a, BASE_JSON/SERVICE_JSON, k t qu	NVS agent_state; c„ nh nh t •ng th ch d li u c– trong myPrefs.
GNSS receiver	C u h nh ri ng trong receiver; c n SAVECONFIG/CFG-CFG v ki m tra sau m t ngu…n.
CONFIG RESET	X„a/n p l i myPrefs v kh•i t o theo needReset; kh ng b o m x„a agent_state ri ng ho c tr ng th i tr n backend.

Serial m i c„ d ng esp_ + ph n nh danh MAC; ID LP_ c– “c Ži ti n t v gi suffix. Ch” l u tr ng th i agent khi identity ƒ n p; token “c ki m tra nh d ng t i firmware, backend vˆn ch u tr ch nhi m x c th c token thi t b th c t .

B ng ph•n v‰ng 4 MB hi n c„ app0/app1 m’i 0x140000 byte, agent_state t i 0x3D0000 k ch th c 0x20000 v coredump t i 0x3F0000. C„ hai slot ng d ng kh ng c„ ngh‡a firmware ƒ tri n khai l nh OTA; danh m c l nh hi n kh ng c„ c p nh t firmware qua m ng.

Quy tr nh c p nh t: ghi l i d ch v /serial, d‰ng †ng board v partition, build r…i upload qua USB, ki m tra log boot v nh n di n chip, tri n khai l i d ch v n u c n, th c hi n nghi m thu. Kh ng d‰ng erase-flash nh b c s a l’i m c nh v c„ th m t identity/license/c u h nh.

```
pio run -e tdm240x_dev_general
pio run -e tdm240x_dev_general -t upload
pio device monitor -b 115200
pio run -e tdm240x_dev_general -t compiledb
```

Ch y t root d n firmware. Trong repo CORS c„ th ch y • root CORS qua c u h nh wrapper ho c v o geodetic/esp32. compiledb c p nh t IntelliSense sau khi th m/t ch file; compile_commands.json ch a €ng dˆn theo workspace, kh ng d‰ng chung nguy n byte gi a c c v tr checkout.

C c env debug/test b t PROGRAM_DEBUG/DUMP_AT_COMMANDS n n log c„ th ch a n i dung ATG/traffic modem. Chia s¢ log ƒ che d li u truy c p; t i li u v v d kh ng ch a m t kh‹u v n h nh th c t .

15. Ki…m tra ph n m m v nghi m thu n i b

```
c++ -std=c++17 -fsanitize=address,undefined -g -Itest/host -Iinclude \
  test/host/base_checks.cpp src/functions/gnss_cmd_builder.cpp \
  src/functions/ublox_commands.cpp src/functions/unicore_commands.cpp \
  -o /tmp/esp32-base-checks
/tmp/esp32-base-checks
python3 -B test/host/check_cors.py /path/to/cors /tmp/esp32-base-checks
python3 -B test/host/check_gnss_stream.py
python3 -B test/host/check_rover_gga.py
```

, ƒ ki m tra build env tdm240x_dev_general, b t o l nh hai protocol/routing v i backend, parser khung xen k£ v GGA m i/c–. , •y l ki m tra host/build, kh ng ph i th nghi m RF ho c UART tr n receiver th t. check_status_flash.py m• cŽng Serial quan s t log sau khi n p; script kh ng upload firmware. C n thi t b th t v Serial Monitor „ng, ki m tra c u h nh DTR/RTS theo board tr c khi ch y.

Bi n b n th c a ph i ghi: board/model/firmware receiver, ngu…n, cŽng/baud, anten v chi u cao, h t a , mode, th€i gian ch€ v s mˆu, account/mountpoint ƒ che th ng tin truy c p, t c /tuŽi RTCM, k t qu rover t i m c v t nh tr ng sau reboot/m t m ng. Ch” x c nh n t sau khi c„ s li u v ng €i nghi m thu.

T i li u tham chi u

- u-blox ZED-F9P Integration Manual: nguy n t‘c survey-in, fixed v cŽng receiver; c n ch n †ng phi n b n t i li u cho thi t b th c t .
- H ng dˆn t ch h“p CORS, command handler, b t o l nh theo chip v c c b i ki m tra host trong c‰ng b n ngu…n l c• s• ki m tra h nh vi firmware.